PRODUCT CATALOGUE

SmartLoop-4

Analogue Addressable Fire Detection & Alarm System

PINPOINT LOCATION **Analogue addressable.** Every detector, call point and module has its own address. In fire the panel shows **exactly which device** - loop, address and location text, for example **LOOP 2 ADDR 017 RELAY ROOM - RACK 3 LEFT**.



SmartLoop-4 Control Panel

4 Loop Analogue Addressable Fire Alarm Control Panel

- 4 loops x 128 devices = **512 analogue addressable devices**
- Designed to **RDSO/SPN/217/2025 Version 3.1** for S&T installations
- Heat & smoke multi-sensors, call points, input and relay modules on the same 2-wire loop
- Built-in isolator in every detector and call point
- 7.0-inch 800 x 480 colour TFT with capacitive touch, key switch, loop LEDs
- 512 fire events + 512 other events with date and time
- 4 action levels, two-detector coincidence, suppression release delay 0-10 min
- TCP/IP to Zonal HQ, RS-485, USB, optional GSM SMS alarms
- 24 h standby + 15 min alarm on battery

Why SmartLoop-4

Finds the fire at once

Loop, address, device type and 32-character location on the display.

Made for railway rooms

Relay rooms, power equipment, battery, ASM, DG, CCTV and OFC rooms.

Fewer false alarms

Smoke + heat sensing, re-check before alarm, dust compensation, coincidence pairs.

Keeps working

Class A loops with isolators; degrade mode if the CPU fails.

Quick to commission

Auto-learn finds every device by its 32-bit serial number.

Stays informed

Network to Zonal HQ, PC software and SMS to railway staff.

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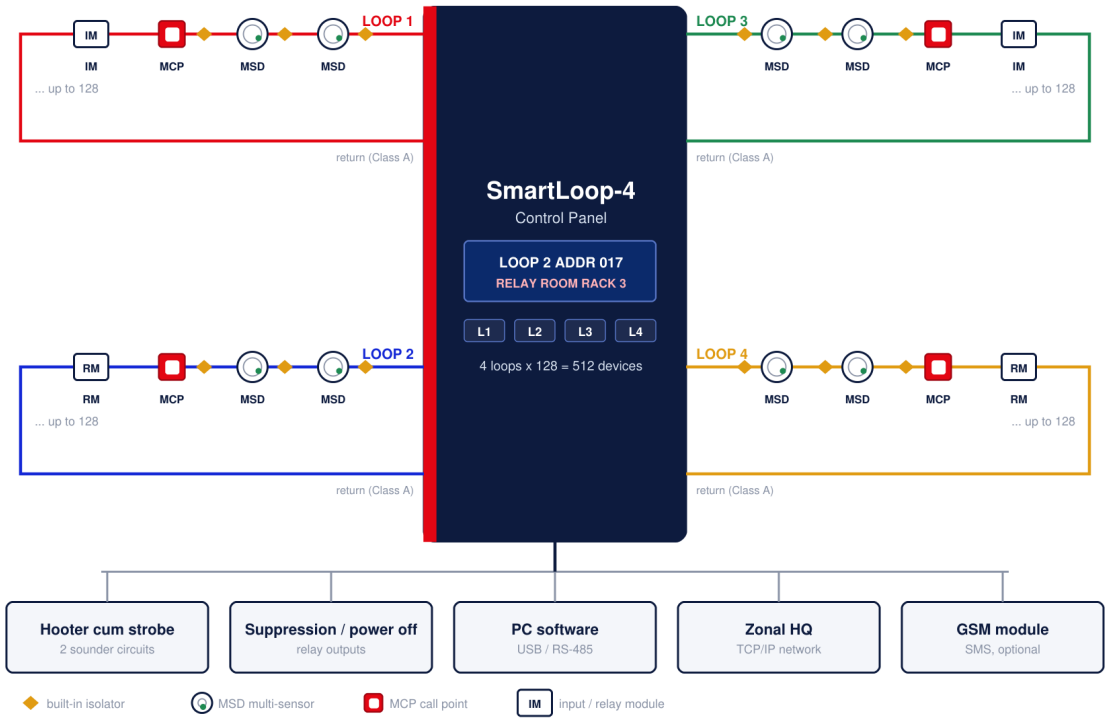


Figure 1 SmartLoop-4 system: 4 loops, addressable devices, outputs and communication

How the panel acts on an alarm

Level	Started by	Panel action
Level 1	One detector of a coincidence pair, or aspirating detector stage 1	Panel buzzer and display; FIRE LED flashing
Level 2	Aspirating detector stage 2	Level 1 + hooter cum strobe in the Station Master room
Level 3	Manual call point, or aspirating detector stage 3	All hooters and strobes; power shut-down outputs (if programmed)
Level 4	Both detectors of a coincidence pair, aspirating detector stage 4, LHS / LHD module	All hooters and strobes within 10 s; suppression release after the set delay (0 to 10 min)

Action levels follow RDSO/SPN/217/2025 Version 3.1 cl. 4.9.16. Delays, rooms and outputs are programmed at the panel or from the PC software.

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The range

ES-MSD100 Heat & Smoke Multi-Sensor



Analogue addressable multi-sensor for relay rooms, power equipment rooms, battery rooms, ASM rooms, DG rooms, CCTV and OFC equipment rooms.

- **Smoke sensing:** Photo-electric (optical) chamber
- **Sensitivity:** 4 levels set from the panel: High 1.5, Normal 2.5 (default), Low 3.5, Very low 4.0 % obs/ft
- **Directional response:** Ratio of highest to lowest response threshold value 1.6 or less
- **Heat sensing:** Fixed temperature 58 deg C and rate-of-rise 6 to 11.1 deg C/min, independent of the smoke sensing
- **False alarm rejection:** Every alarm checked again before it is sent

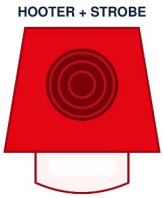
ES-MCP100 Manual Call Point



Addressable manual call point for entrances, exits and escape routes.

- **Operation:** Push the resettable element
- **Indicator:** Green blink = healthy
- **Isolator:** Built-in short-circuit isolator

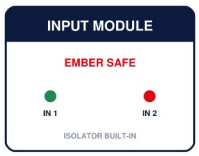
ES-HS100 Hooter cum Strobe



Combined audio and visual alarm for the Station Master room, relay room and ESM duty room.

- **Hooter:** Piezo-electric
- **Strobe:** Flashing red / white
- **Connection:** Panel monitored sounder circuit, 24 V DC

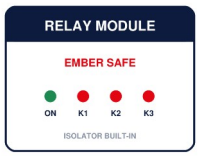
ES-IM100 Input Module



Brings other fire detection equipment onto the SmartLoop-4 loop as addressable points.

- **Inputs:** 2 monitored inputs (open / short / active), each with its own address
- **Use:** LHS cable or LHD interface, aspirating smoke detector stages, UV / IR flame detectors, probe-type bi-metallic heat detectors, suppression system feedback
- **Isolator:** Built-in short-circuit isolator

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ES-RM100 Relay Output Module

Addressable outputs for automatic fire suppression, power shut-down and the data logger.

- **Contacts:** 3 NO + 3 NC potential-free contacts, at least 500 mA each
- **Control:** Programmable from the panel: action level, delay (0 to 10 min in 0.5 min steps), room
- **Isolator:** Built-in short-circuit isolator



ES-GSM100 GSM Module (optional)

SMS alarm messages to railway staff.

- **Network:** Quad-band GSM 850 / 900 / 1800 / 1900 MHz
- **Messages:** SMS to 5 or more programmable mobile numbers for action levels 2, 3 and 4, within 30 s
- **Supply:** From the panel

Ordering information

Code	Product	Description
ES-SL4	SmartLoop-4 Control Panel	4 loop analogue addressable fire alarm control panel, 512 addressable devices
ES-LC128	Loop Card	Plug-in loop controller card, 128 analogue addressable devices (one per loop)
ES-MSD100	Heat & Smoke Multi-Sensor	Analogue addressable optical smoke + fixed / rate-of-rise heat detector with built-in isolator
ES-MCP100	Manual Call Point	Addressable, resettable manual call point with built-in isolator
ES-HS100	Hooter cum Strobe	Piezo-electric hooter, up to 90 dB at 1 m, with red / white flashing strobe
ES-IM100	Input Module	Addressable monitored input module for LHS / LHD interfaces, aspirating detectors, flame and probe heat detectors, suppression feedback
ES-RM100	Relay Output Module	Addressable relay module, 3 NO + 3 NC potential-free contacts, for suppression release, power shut-down and data logger
ES-GSM100	GSM Module (optional)	Quad-band GSM modem: SMS alarms to 5 or more mobile numbers
ES-SLPC	SmartLoop PC Software	Windows software: site layout view, event download and analysis, configuration
ES-BAT	Standby Battery	2 x 12 V sealed maintenance-free lead-acid batteries, sized for 24 h standby + 15 min alarm

The SmartLoop-4 also connects, through ES-IM100 input modules, the LHS cable / LHD interfaces, aspirating smoke detectors, UV & IR flame detectors and probe-type bi-metallic heat detectors called for in RDSO/SPN/217/2025 Version 3.1.

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